

The Law of Fairness: Terminal neutrality as a boundary condition on conscious state space

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Abstract

TL;DR: The Law of Fairness hypothesizes that every conscious life's cumulative emotional balance integrates to exactly zero at the termination of that conscious stream, not as karma but as a testable physical constraint on conscious state space. The model formalizes a latent life ledger $L(T) = \int_0^T F(s) ds$ representing time-integrated affective load and requiring that admissible trajectories satisfy the boundary condition $L(T) = 0$. This closure arises from a multiplicatively coupled stochastic process $L(t) = U(t)Y(t)$, where declining biological reserve $U(t)$ compresses stochastic fluctuations in $Y(t)$. As reserve approaches collapse near the stopping time T , the diffusion term shrinks and the model predicts terminal variance compression $\lim_{t \rightarrow T} \text{Var}(L(t)) = 0$. Because the endpoint T is defined through an independent biological signal rather than the ledger itself, the prediction avoids tautology and becomes empirically falsifiable. The framework therefore stands or falls on observable signatures in physiological telemetry and longitudinal affect data and is presented as a formal hypothesis for the academic community to test and attempt to falsify.

The Law of Fairness (LoF) is presented as a formal, falsifiable framework. It invites rigorous empirical scrutiny and cross-disciplinary validation according to the standard methodologies employed in biophysics, systems neuroscience, and mathematical biology.

The validity of the LoF depends upon its ability to withstand robust falsification protocols. If it is true, it leaves constraint signatures that are mathematically impossible to reproduce with ordinary homeostasis, hedonic adaptation, or ensemble-based reinforcement learning. The framework therefore treats fairness not as a moral ideal but as a candidate physical constraint on the trajectory of conscious state space.

Each proposed mechanism in this framework is motivated by published findings across affect dynamics, sleep physiology, allostatic energetics, horizon-dependent valuation, and inhibitory control. The theoretical scaffolding is locked, the empirical alignments are explicit, and the preregistered falsifiers are public. Ultimately, the framework's utility will be determined by prospective data and out-of-sample predictive accuracy.

I. The Core Hypothesis & Mathematical Framework

To eliminate semantic ambiguity, we define the parameters strictly:

- **F(t)**: instantaneous net affect / valence rate (latent).
- **z_k(t)**: preregistered intensive, non-conservative physiological rates (e.g., ATP-equivalent metabolic expenditures).
- **HCI(t)**: Hedonic Composite Index; the preregistered empirical estimator built from z_k(t).
- **L(t) = ∫₀^t F(s) ds**: latent cumulative ledger.
- **Ĺ(T) = Σ HCI(t_i) Δt_i**: measured ledger estimator.
- **θ(t)**: Unity Index (orthogonal proxy for conscious access unity, e.g., perturbational complexity indices; Casali, 2013).
- **T**: endpoint stopping time (Unity Index threshold crossing).
- **U(t)**: independently measured reserve/plasticity proxy.
- **H(t)**: remaining conditional horizon estimate.
- **Φ**: compensability score / future-preserving admissibility weight.
- **λ(t)**: shadow price / Lagrange multiplier weighting compensability as horizon collapses.

The Law asserts exact terminal neutrality at the end of the unified stream. In its strong form, it asserts a path constraint rather than an ensemble tendency: $P(L(T) = 0) = 1$ in the latent process, subject to empirical approximation where $|\dot{L}(T)| \leq K$ accounts for proxy uncertainty. A unified conscious life is a single, time-irreversible, non-ergodic path terminating at an absorbing boundary.

Multiplicative Coupling and Itô Dynamics To avoid mathematical tautology, the ledger is multiplicatively coupled to the biological Unity Reserve $U(t)$, representing residual epigenetic and metabolic plasticity. $U(t)$ decays toward zero ($dU(t) = -v(t) dt$). Let $Y(t)$ be an unconstrained diffusion process defined by $dY(t) = \sigma dW(t)$ with an arbitrary initial state $Y(0) = Y_0$. The coupled ledger is defined by the product representation: $L(t) = U(t) Y(t)$

Applying Itô's Lemma yields the governing dynamics (including the cross-variation term):
 $dL(t) = -(v(t)/U(t)) L(t) dt + \sigma U(t) dW(t) + \sigma \gamma \rho dt$

As $U(t) \rightarrow 0$ near the endpoint, two critical empirical signatures emerge:

- **Drift Dominance:** The mean-reversion drift term $v(t)/U(t)$ diverges, forcing rapid, inescapable convergence toward zero.
- **Variance Compression:** The diffusion coefficient $\sigma U(t)$ vanishes, suppressing stochastic excursions and producing mandatory variance compression.

These dynamics generate superlinear horizon weighting and aggressive pruning of high-variance trajectories via the Queue System (QS) as the conditional horizon $H(t)$ shrinks.

II. The Endpoint Firewall & Statistical Rigor

Empirical validation must prioritize the assessment of the terminal boundary condition. “Death of Mind” is defined operationally as a causal stopping time driven by a preregistered Unity Index threshold, not by somatic death. Formally, $T = \inf \{ t \geq 0 : \theta(t) \leq \theta_0 \}$, with the event $\{T \leq t\}$ measurable with respect to the filtration $\mathcal{F}_t$. If you define “death” as “the time the ledger hits zero,” then neutrality is a tautology. The LoF framework explicitly precludes this definition to ensure the independence of variables. The Unity Index $\theta(t)$ must be derived from physiological channels strictly orthogonal to the HCI to prevent statistical circularity.

The Telescoping Hazard: If physiological telemetry relies on exact, conservative state variables, the Riemann sum intrinsically telescopes to $S(T) - S(0)$, rendering the path irrelevant. To prevent algebraic collapse, LoF mandates that empirical observables must be non-conservative, path-dependent thermodynamic rates (e.g., allostatic wear, continuous ATP consumption per the Energetic Model of Allostatic Load; Bobba-Alves, 2022). Terminal neutrality is hypothesized as a dynamical outcome of the system's evolution rather than an a priori algebraic identity.

III. Empirical Domains & Falsification Protocols

Before diving into the lab work, here are the unique predictions that separate LoF from standard models:

- Path-wise closure at a strictly state-coupled (not exogenously random) stopping time.
- Mandatory variance compression scaling strictly with a measured biological collapse proxy.
- A specific horizon-sensitive compensability weighting predicting inhibitory-braking signatures in the brain.
- A mechanistic REM inversion channel functioning as an offline thermodynamic counterweight.

In-Silico Falsification: The Virtual Terminal Maze Imagine a computer-simulated “rodent” subject to severe allostatic debt placed in a virtual maze with 100 exits. 99 exits lead to death (rigged with misleading, high-arousal lures), and 1 exit leads to survival. Under standard Reinforcement Learning, the agent follows the immediate utility of the lure and perishes. Under the LoF non-ergodic controller, as the horizon $H(t)$ hard-caps and $U(t)$ approaches zero, the shadow price of compensability ($\lambda(t)$) skyrockets. The controller must aggressively brake against the lures. The strict prediction is that despite adversarial cues, the success rate will significantly exceed unconstrained baselines due to the spiking shadow price of compensability.

Domain 1: The Queue System & Admissible-Set Pruning In cognitive labs, horizon-scaled Φ must explain variance in valuation and control hubs beyond standard predictors (utility, conflict, arousal). Anchored in the Expected Value of Control framework (Shenhav, 2013), the right inferior frontal gyrus (rIFG) and dACC aggressively brake low-compensability choices. Admissible menu counts must decrease proportionally to $H(t)^{-1}$ and exhibit overdispersion rigorously tested via preregistered Negative Binomial generalized linear mixed models. If disabling this circuitry via TMS/tDCS does not produce admissible-set leakage, the mechanism fails.

Domain 2: Systems Biology & The Thermodynamic Cost Unresolved negative valence (high variational free energy) is a measurable drain on ATP. High-variance trajectories systematically accelerate cellular epigenetic aging under the Energetic Model of Allostatic Load (Juster, 2010), serving as the physical substrate of $U(t)$ decay. If the subjective ledger drifts into permanent deficit without accelerating the thermodynamic collapse of $U(t)$, the physical anchoring is broken.

Domain 3: Horizon Scaling & Neural Revaluation As the biological horizon collapses, the vmPFC must encode a distinct value surplus specifically for highly compensable, reparative choices. We predict a strict $\Phi \times H(t)^{-1}$ interaction in the BOLD/EEG signal.

Domain 4: Sleep Physiology & Noradrenergic Blockade When waking life offers no behavioral path to balance, LoF predicts a compensatory shift toward more positively valenced or mastery-themed states during healthy REM sleep (extending Cartwright, 1998). Mechanism: normal noradrenergic suppression allows affective reweighting without autonomic stress. Caveat & Falsifier: REM's noradrenergic suppression is documented to fail in PTSD-like physiology (Germain, 2008). This is a quantifiable boundary: if recurrent pathological failures prevent this inversion at a population prevalence exceeding the preregistered measurement error bound K , the 100% guarantee is definitively falsified. While hypothesized as a modifiable vulnerability factor, bidirectional causality between PTSD and

sleep disturbances is acknowledged; preregistered longitudinal designs will disentangle directions via cross-lagged models.

Domain 5: Social Coupling & Scarcity The framework predicts an emergent shadow price on scarce relief opportunities, prioritizing those nearer closure. If individual behaviors do not synchronize under shared scarcity, universality fails.

Domain 6: Gerontology & Terminal Variance Compression If the Unity Reserve is collapsing, physiological flexibility (HRV) collapses with it, and the cross-sectional ledger distribution must contract. Neutrality is corroborated only if both one-sided tests reject the null, meaning the 95% confidence interval of the measured estimator $\hat{L}(T)$ lies entirely within $[-K, +K]$. To prevent subjective tuning, TOST is supplemented with Bayes factors computed against a preregistered uninformative prior. $BF_{01} > 30$ (very strong evidence) corroborates neutrality, and $BF_{10} > 30$ favoring terminal imbalance constitutes a definitive statistical falsification of the primary hypothesis.

IV. The Meta-Level Hypothesis: State-Dependent Reactions

Methodological Controls and Interpretive Constraints: This is an auxiliary prediction. If it fails, it does not rescue the core LoF; it merely prunes one extension. Strong negative reactions can still be correct, and enthusiastic acceptance can still be mistaken.

A self-referential prediction of the LoF is that an individual's reaction to the hypothesis itself is not purely a rational judgment. It functions as a biological output modulated by their current latent affective ledger state, $L(t)$. When a person encounters the strong-form LoF hypothesis, their internal generative model simulates the imposition of this terminal boundary condition. If their current $|L(t)|$ is extremely high (either a massive negative deficit like chronic unresolved pain or a massive positive surplus like unearned hedonic excess), the projected metabolic cost of restoring balance triggers immediate defensive pruning of the idea itself via the Queue System. The shadow price of compensability, $\lambda(t)$, skyrockets, and the system actively suppresses engagement with the hypothesis to protect its trajectory.

Reaction Profiles:

- **Massive Deficit:** The hypothesis feels existentially threatening because it reframes suffering as part of an inevitable thermodynamic balancing process. Defensive rejection is common.
- **Massive Surplus:** The LoF is perceived as an imposed future compensatory cost. Existential dread or defensive pathologizing follows.
- **Near-Neutral (High HRV):** The hypothesis poses minimal immediate threat. Reactions tend toward intellectual curiosity.

Empirical Test: This meta-hypothesis is strictly falsifiable. The central prediction is a positive correlation between absolute distance from neutrality and aversive reaction magnitude: $E[|R(t)|] = \alpha + \beta_1 |\dot{L}(t)| + \beta_2 g(H(t)) + \beta_3 h(U(t)) + \beta_4 |\dot{L}(t)| g(H(t))$

Load-Bearing Beliefs and Paradigm Shifts Every person relies on central load-bearing concepts (religious faith, scientific worldview, etc.). Within the Free Energy Principle (as detailed in the manuscript's FEP mappings), these function as high-precision priors. If a new concept threatens one of these priors, it generates a cascade of prediction errors. The metabolic cost (ATP expenditure) of rebuilding that global model is thermodynamically prohibitive. The Queue System pre-emptively prunes the threatening idea to avoid an allostatic collapse.

Ethical Guardrail: This construct must never be used to dismiss criticism. Strong reactions are data about constraint engagement, not evidence of ignorance.

V. Implementation and Future Research Directions

Preregistration packages, HCI code templates, power-analysis scripts, and ethical templates are being prepared (see the full manuscript text below for detailed stochastic derivations and protocols, or the GitHub repository for resources). Red-team bounties will be posted for adversarial fits and null results.

Quickstart Falsification Tests (No New Equipment Needed):

- **Terminal Variance Compression (Hospice):** Fit affect variance vs. time-to-T. Preregister that variance must contract as a function of the Unity proxy.
- **Horizon × Compensability (Decision Tasks):** Preregister a $\Phi \times H(t)^{-1}$ interaction predicting choice signals.
- **REM Inversion Channel (Sleep Labs):** Test if high negative waking load predicts next-night REM affective reweighting.

Model Parsimony and Comparative Testing: If a parsimonious adversarial model, utilizing only standard homeostatic regulation and ordinary memory consolidation, yields equivalent or superior out-of-sample predictions of terminal behavior and horizon effects, the LoF framework should be rejected in favor of the simpler alternative.

Reply here or reach out to coordinate. Tell us your lab's expertise, and we will match you to the exact protocol. The question is no longer philosophical; it is strictly empirical. The scientific utility of this hypothesis rests not on its theoretical novelty, but on its exposure to rigorous empirical falsification.

Law of Fairness: Mathematical Formalization and Stochastic Dynamics

The Law of Fairness (LoF) posits a hypothesized universal boundary condition on the trajectory of any unified conscious stream, remaining a testable conjecture awaiting empirical confirmation across diverse populations. While hypothesized as universal, initial empirical tests will focus on human populations, with extensions to non-human conscious systems (e.g., via animal models) requiring separate validation. Let $F(t)$ represent the net valence rate (affect) at time t . The latent life ledger is defined as the time integral of this rate over the conscious lifespan $[0, T]$:

$$L(T) = \int_0^T F(s) ds$$

Strong Form LoF (hypothesized boundary condition): for every unified conscious stream, the latent terminal ledger satisfies $L(T) = 0$ in the underlying process. $P(L(T)=0)=1$ in the latent process. Empirically, we test whether the estimator $\hat{L}(T)$ falls within a preregistered equivalence band $|\hat{L}(T)| \leq K$ given measurement uncertainty. This is not a regression to the mean; it is a strict path-wise closure constraint on admissible trajectories rather than an ensemble expectation.

The Shadow Price of Admissibility: terminal neutrality functions as a binding Lagrangian constraint on the action integral of the conscious state space. Within the Free Energy Principle framework, uncompensated affective imbalance at the collapse of the conscious workspace corresponds to unresolved prediction error. Trajectories violating the terminal boundary are assigned zero probability under the model's imposed constraint.

Epistemological Framework (Testing): because $L(t)$ and T are latent quantities, empirical tests rely on preregistered operational proxies acting within a formal state-space observation model. We approximate the valence rate using a Hedonic Composite Index (HCI) constructed from longitudinal physiological telemetry. Crucially, these physiological observables (e.g., heart rate variability, endocrine markers) are not assumed to be literal, direct energy meters (calorimetry); rather, they function as hypothesized estimators for the latent regulatory burden, akin to hidden state estimation in Kalman filtering or computational neuroscience. While HCI provides objective telemetry, it approximates subjective valence; preregistered experience sampling methods (ESM) must be integrated as a secondary validator. Agreement between HCI and ESM must exceed a preregistered minimum reliability threshold $\kappa \geq 0.6$ (or $ICC \geq 0.6$ for continuous scales), representing moderate agreement in standard psychometric benchmarks. Values below this threshold trigger proxy recalibration or hypothesis rejection. Multi-modal fusion (e.g., Bayesian integration of fMRI/ESM/HRV) will quantify uncertainty, and failures in proxy invariance falsify the operationalization itself.

Valence polarity is hypothesized to correspond approximately: positive $F(t)$ denotes low free energy (eudaimonia), while negative valence corresponds to high free energy (dysphoria) requiring metabolic work. This hypothesizes valence as a phenomenological correlate of variational free energy, extending speculative mappings in FEP (Smith, 2019; Hesp, 2021), which remain theoretical and require empirical validation via preregistered studies linking HCI to prediction error proxies (e.g., via EEG mismatch negativity), though this link is theoretical and awaits direct causal testing via interventions targeting prediction error (e.g., pharmacological modulation of mismatch negativity or optogenetic modulation in animal models). This mapping is provisional and will be falsified if preregistered interventions fail to show HCI changes correlating with FE proxies (TOST 90% CI within $[-0.4, 0.4]$ or $BF_{10} < 1/3$ favoring no association; $n \geq 100$ per arm powered for 80% detection of $r \geq 0.4$).

Mappings ensure $W \geq 0$, incorporating sensitivity tests for asymmetric costs. Negativity bias is modeled via asymmetric weights (e.g., $W_{neg} > W_{pos}$, calibrated from preregistered ESM data), testing $L(T)$ robustness under varying bias ratios (e.g., 3:1 to 5:1, calibrated from domain-specific meta-analyses like Baumeister, 2001 for general negativity, and Gottman, 1994, 1999 for relationships), with preregistered tests exploring cultural moderators (e.g., individualism scores per Hofstede, 2010; falsify if moderator effects explain $< 10\%$ variance, $R^2 < 0.1$).

Continuous digital phenotyping transforms static level measurements (e.g., cortisol baselines, HRV windows) into intensive metabolic expenditure rates via preregistered phenomenological mapping functions. These mappings assume approximate proportionality under preregistered calibration, acknowledging potential non-linearities tested via sensitivity analyses (including non-parametric models like GAMs, with rejection if non-linearity yields better fit per AIC difference > 2 or explains $> 15\%$ additional variance via partial R^2), including subject-specific modeling (e.g., via mixed-effects regressions incorporating genetic covariates like BDNF polymorphisms) to ensure proxy validity across heterogeneous populations (e.g., bootstrapped variance estimates, modeling HRV variance as proportional to ATP-equivalent costs, extending models of allostatic load energetics (Juster, 2010; Bobba-Alves, 2022). Future work must calibrate these mappings via bioenergetic assays. This translates affective dynamics into measurable biological costs (McEwen & Wingfield, 2010).

We model $F(t)$ as proportional to the instantaneous rate of change in cumulative allostatic wear, written heuristically as $F(t) \approx -W(t)(dS/dt)$ within a continuous dynamical framework where S denotes cumulative allostatic wear rather than a conservative potential, associated with deviation from physiological or control balance, and $W \geq 0$ is a preregistered scaling constant linking state deviation to energetic cost. A central structural concern is that if affect

were modeled strictly as $F(t) = -W(dS/dt)$, then $\int_0^T F(s) ds$ collapses to $-W \int_0^T (dS/dt) dt = -W(S(T) - S(0))$. If one assumes $S(T) = S(0)$ by construction, neutrality becomes a boundary identity rather than a dynamical outcome. LoF does not assume $S(T)=S(0)$. The integral permits non-neutral outcomes in principle. This ensures that the observables used to construct $F(t)$ are not gradients of a conservative state variable; if they were conservative, the integral would collapse to $S(T)-S(0)$ by the fundamental theorem of calculus. For example, if $F(t)=1$ on $[0, T]$, then $L(T)=T \neq 0$. Therefore, the claim $L(T)=0$ is empirically falsifiable rather than analytically guaranteed.

The Telescoping hazard: If physiological telemetry relies on exact conservative state variables, the Riemann sum intrinsically telescopes to $S(T) - S(0)$, rendering the path irrelevant.

Methodological Note on Integration: To prevent algebraic collapse, the Law mandates that the empirical observables z_k cannot be conservative state variables. They must represent non-conservative, path-dependent thermodynamic expenditures such as cumulative allostatic wear or continuous ATP consumption, as quantified in models of allostatic load energetics (Bobba-Alves, 2022). Because empirical measurements arise from discrete physical telemetry, the empirical estimator is constructed as a discrete Riemann sum on these path-dependent rates. To mitigate discretization errors and ensure non-conservativeness, $z_k(t_i)$ must include irreversible metabolic expenditure proxies (e.g., heat dissipation proxies from wearable calorimetry).

Preregistered pipelines incorporate stochastic noise models (e.g., telemetry variance) and Monte Carlo simulations ($n=1000$) to confirm path-dependence ($\text{Var}(\dot{L}) > 0$ for non-constant paths). Preregistered tests will use path-dependence diagnostics and trajectory permutation tests to confirm $z_k(t_i)$ does not integrate to a state difference (i.e., the cumulative integral cannot be expressed as $S(T) - S(0)$) ($p < 0.05$ rejecting the null hypothesis that the cumulative signal can be represented as a conservative state difference $S(T) - S(0)$, with preregistered power analysis targeting 80% detection of small effects, $d=0.2$, $n \geq 788$ for two-tailed t-test):

$$HCl(t_i) = \sum w_k z_k(t_i)$$

where $z_k(t_i)$ are preregistered intensive rates (e.g., instantaneous ATP consumption derived from allostatic markers, preventing Δt cancellation and ensuring path-dependence per non-conservative thermodynamics).

The empirical ledger becomes: $\dot{L}(T) = \sum HCl(t_i) \Delta t_i$ (where $z_k(t_i)$ represents the intensive physiological rate, with z_k as non-integrable rates, e.g., entropy production, preventing Δt cancellation).

Structural safeguards:

- **Preregistration:** Weights w_k , anchors, and preprocessing pipelines are fixed prior to data collection.
- **Invariance:** LoF assumes metric invariance of valence across demographics. Preregistered tests (e.g., multi-group confirmatory factor analysis on HCI) must confirm loading equivalence (TOST 90% CI within preregistered bounds or $BF_{10} < 1/3$ favoring no difference) in diverse global cohorts (e.g., non-WEIRD samples), incorporating cultural covariates (e.g., individualism scores per Hofstede, 2010). Failures restrict claims to validated subpopulations and trigger adaptive recalibration.
- **No post hoc tuning:** adjustments to w_k after observing outcomes are prohibited. Neutrality must emerge from system dynamics rather than algebraic identity.

Defining “Death of Mind” via retrospective irreversibility requires future information, which violates real-time causality. Instead, the endpoint T is defined as a stopping time with respect to the filtration $\mathcal{F}_t$:

$$T = \inf \{t \geq 0 : \theta(t) \leq \theta_0\}$$

here, $\theta(t)$ is a preregistered proxy for the unity of conscious access, anchored to markers such as perturbational complexity and cortical integration indices (Casali, 2013), adapted non-invasively via EEG approximations (Comolatti, 2019) and extended from disorders of consciousness research to monitor degradation in healthy trajectories. The event $\{T \leq t\}$ is $\mathcal{F}_t$ -measurable.

The endpoint is defined independently as the stochastic stopping time $T = \inf\{t \geq 0 : \theta(t) \leq \theta_0\}$. The observable proxy $\theta(t)$ is assumed to be a non-increasing indicator on average of the latent biological reserve $U(t)$, with fluctuations modeled via stochastic volatility. Checks will validate decreasing monotonicity via trend tests (e.g., Mann-Kendall, $\tau < -0.5$) on longitudinal EEG data from healthy cohorts (e.g., extending UK Biobank studies), with violations ($\tau > -0.5$) prompting semi-parametric modeling (e.g., GAMs) or restriction to subpopulations where monotonicity holds (validated via subgroup analysis), so that collapse of $\theta(t)$ operationally corresponds to depletion of $U(t)$. Death of mind corresponds to the collapse of unified experience capacity (represented by θ), not by the content of the experience captured by L .

Ledger gating is explicit. Define the unity gate: $g_\theta(t) = 1_{\{\theta(t) > \theta_0\}}$

The ledger accrues only while the gate is open: $L(t) = \int_0^t g_\theta(s) F(s) ds$

This blocks post-collapse integration and cleanly separates the endpoint definition from the ledger value.

The Logical Fork Lemma: If T is independent of the sigma algebra generated by the ledger process prior to state coupling, and the conditional distribution of $L(T)$ admits a non-degenerate absolutely continuous density, then $P(L(T)=0)=0$. The lemma assumes non-degenerate absolutely continuous density and independence, under which exact neutrality has measure zero under exogenous stopping. For affective dynamics featuring discrete events (e.g., trauma), this extends to jump-diffusion models (Merton, 1976), with preregistered simulations (including jump intensities calibrated from preregistered ESM data, e.g., variance decomposition showing jumps explain 10-20% of HCI fluctuations, $\lambda=0.1-0.5$ jumps/day, size $\sim N(0,1)$) testing $P(L(T)=0)$ robustness via 10,000 Monte Carlo runs. For a diffusion with continuous density, the probability that the random variable equals any exact value is zero unless the distribution contains a point mass.

To satisfy the ontological guarantee $P(L(T)=0) = 1$, at least one lemma condition must fail (e.g., via terminal concentration in the multiplicative model $L(t) = U(t)Y(t)$). Closure cannot arise from randomness alone without state-coupling or degeneracy. LoF therefore satisfies this via terminal degeneracy ($U(t) \rightarrow 0$), committing to state-coupled termination. This coupling is consistent with age-related dopamine models compressing affect variance (Li & Sikström, 2002; Bäckman, 2010), and is explicitly testable via Vector Autoregression (VAR) models on HCI and $\theta(t)$. Impulse response functions must show variance compression (effect size > 0.3), extending lags to 1-12 for seasonal effects alongside longitudinal HRV-ledger correlations. The hazard of stopping depends on the ledger state or its derivatives, representing vanishing diffusion with boundary-attracting dynamics near the terminal boundary.

Edge Cases and Abrupt Termination: For abrupt, exogenously forced terminations (e.g., sudden accidents) where $U(t)$ does not decay continuously, the framework predicts preemptive balancing via hypothesized anticipatory stress markers (e.g., altered autonomic activity), testable via retrospective HCI trends in accident cohorts; note that Chida & Steptoe (2009) supports prospective cortisol-mortality links for cardiovascular events but requires extension to non-CV abrupt cases via retrospective cohorts with ambulatory monitoring. This is falsifiable if $>15\%$ (aligned with Chida & Steptoe, 2009 meta-prevalence of 15-25% in CV cohorts, extended to non-CV via preregistered ambulatory data; powered for 80% detection, $n \geq 153$ per cohort assuming binomial test) of abrupt cases show $|\dot{L}(T)| > K$ without these markers.

For comas or severely truncated lifespans (e.g., infants), LoF predicts the unity gate prevents accrual without balancing. For flickering $\theta(t)$, a smoothed gate (e.g., 7-day moving average,

validated if it reduces artifactual variance by >50% in simulations) is used to avoid artifactual accrual. This validates the gate $g_{\theta}(t)$ via developmental EEG studies showing $\theta(t)$ thresholds align with consciousness onset.

A frequent objection from computational disciplines is that reinforcement learning or homeostatic regulation already explains affective dynamics. The limitation is ergodicity. Standard reinforcement learning optimizes expected cumulative reward over an ensemble of trajectories: $\arg \max E[\int_0^T r(t) dt]$. Furthermore, in AI agent evaluation, if a success metric is too tightly coupled to its constraint boundary, the system inevitably engages in "reward hacking," exploiting the definition rather than solving the environment. The Law of Fairness avoids biological reward hacking by mandating that the boundary condition (the escalation threshold, or collapse of the Unity Index) and the policy behavior (the HCI ledger) are measured orthogonally. This ensures that terminal variance compression emerges as a genuine, observable physical constraint signature, rather than a tautological restatement of the stopping rule.

While standard ensemble-optimizing RL permits extreme path-wise outcomes, LoF hypothesizes strict non-ergodic path-wise closure (though subprocess ergodicity within neural populations is possible). This contrasts with standard RL but aligns with safe-RL variants (Garcia & Fernández, 2015). Preregistered simulations (e.g., $n=1000$ trajectories using OpenAI Gym) demonstrate that unconstrained RL permits $|L(T)| > K$ in a non-negligible fraction of simulated trajectories (based on simulation benchmarks for unconstrained RL variance, e.g., via OpenAI Gym baselines, with exact thresholds preregistered post-pilot $n=100$, adjusting threshold if baseline variance differs by >10%), whereas LoF suppresses this via horizon-scaling. Extending cognitive effort models (Shenhav, 2013), this is testable via priced constraint experiments using non-parametric power (e.g., bootstrap) for robustness. It is a single time-ordered trajectory terminating at an absorbing boundary T . In non-ergodic systems, the time average of a realized path does not converge to the ensemble expectation.

Because LoF requires exact path-wise closure $L(T) = 0$, the governing dynamics cannot remain an unconstrained expectation optimizer. As the horizon shrinks, dispersion must be actively suppressed and feasible actions must tilt toward compensable trajectories.

Empirical Test, Priced Constraint: Experimental tasks introduce choices where high-compensability actions carry a small immediate cost. Prediction: as the conditional horizon H shrinks, willingness to incur an immediate cost for closure increases. Models must control for hyperbolic discounting, baseline risk, and loss aversion via preregistered covariates (e.g., Prospect Theory parameters per Kahneman & Tversky, 1979) to isolate Φ effects. The coefficient $\beta(H \times \Phi)$ must be significantly positive under preregistered baselines, with

multicollinearity checked for all terms, including interactions ($VIF < 5$; if violated, use ridge regression with λ selected via cross-validation), and statistical power targeting 80% detection of medium effects ($d=0.5$ for main effects, $d=0.3$ for the interaction term $\beta(H \times \Phi) > 0$, requiring $n \approx 88$ per cell). Extending Shenhav (2013), preregistered fMRI tasks must isolate Φ variance in control hubs via GLM contrasts ($\beta > 0.2$, based on meta-analytic control hub effects), with null results falsifying the mechanism.

We construct a stochastic process that satisfies Strong-Form LoF without mathematical tautology. The Unity Reserve $U(t)$ is defined as residual biological plasticity with a measurable physical substrate. Over a lifespan, predictable epigenetic degradation and metabolic entropy accumulation persistently reduce plasticity. In the CNS, this appears as reduced synaptic plasticity and perineuronal net rigidification. $U(t)$ represents the inverse of cumulative allostatic load. Adapted and of finite variation, we model net plasticity decline but allow small reversible fluctuations:

$$dU(t) = -v(t) dt + \gamma dW_U$$

where $v(t) > 0$ is an adapted stochastic process incorporating physiological noise. To ensure T remains a finite stochastic stopping time, we assume $v(t)$ is bounded below by $\varepsilon > 0$ via a reflected SDE, with γ bounded by $0 < \gamma < \sqrt{2\varepsilon}$ to ensure positive recurrence and $P(T < \infty) = 1$ (validated via Lyapunov functions in simulations). We acknowledge partial reversibility of aging markers via medical interventions (Horvath, 2013), restricting the LoF boundary condition to natural trajectories without interventions reversing $>10\%$ of $U(t)$ decline (per Horvath clock metrics). $T = \inf\{t \geq 0 : \theta(t) \leq \theta_0\}$, where $\theta(t)$ is a preregistered proxy for unity of conscious access, derived from physiological channels strictly orthogonal to the HCI (e.g., EEG perturbational complexity indices vs. autonomic HRV/cortisol markers). To guarantee this, preregistration must include empirical thresholding (e.g., Canonical Correlation Analysis) and mutual information thresholds (< 0.1 nats, chosen as a conservative low dependence threshold based on simulation calibration, e.g., Kraskov, 2004) proving $\theta(t)$ and HCI share negligible mutual information prior to the endpoint, with a fallback to instrumental variable methods with genetic instruments (e.g., BDNF as proxy) if partial dependence is detected.

This definitional mapping is standard in biophysical modeling and does not make the stochastic dynamics irrelevant; the diffusion process still strictly governs the trajectory, the variance compression, and the hitting time. While the multiplicative structure structurally enforces terminal neutrality, it serves as a mathematical proof-of-concept demonstrating that path-wise closure is dynamically achievable without backward causation (algebra enforces closure given empirical $U(T) > 0$; falsify via $\theta(t)$ not correlating with variance compression, $r < 0.3$). Falsification does not test the algebra; it tests whether actual

biological systems exhibit the predicted physical signatures of this constraint (e.g., drift dominance, variance compression). Let $Y(t)$ be defined by $dY(t) = \sigma dW(t)$ with an arbitrary initial state $Y(0) = Y_0$. We define the ledger by multiplicative coupling:

$$L(t) = U(t)Y(t)$$

This coupling mirrors established biology, such as affect amplification in high-plasticity states (e.g., adolescence) versus compression in aging (e.g., via reduced dopamine modulation; Li & Sikström, 2002). Closure follows from the imposed biological decay constraint together with multiplicative coupling. Because T is a finite stochastic stopping time and $Y(t)$ has continuous sample paths, the random variable $Y(T)$ is finite almost surely (if extended to heavy-tailed Lévy processes, $Y(T)$ remains finite almost surely under the moment condition $\alpha > 1$). If the unconstrained process has heavy tails (e.g., an α -stable Lévy process), we restrict the stability parameter α so that the process admits a finite first moment (e.g., $\alpha > 1$), confirming via preregistered tail index estimation (e.g., Hill's method, estimating the tail index α with confidence intervals testing whether $\alpha > 1$ so the first moment exists; if data deviate from α -stable (tested via goodness-of-fit $p < 0.05$), fallback to empirical tail estimation via EVT). Therefore, at the boundary, the stochastic variance of $Y(t)$ is completely absorbed by the zero-state of the biological multiplier, satisfying:

$$L(T) = U(T)Y(T) = 0 \times (\text{finite random variable}) = 0 \text{ identically.}$$

For $t < T$, Itô's lemma yields dynamics:

$$dL(t) = -(v(t)/U(t))L(t)dt + \sigma U(t)dW(t) + \sigma\gamma\rho dt$$

(Here, $v(t)$ is adapted but treated as non-diffusive for simplicity; given $dU = -v dt + \gamma dW_U$, we add the Itô cross-term $\sigma\gamma\rho dt$. We estimate ρ via Spearman correlation (for robustness to non-normality) on paired $HCl-\theta(t)$ data; if $|\rho| > 0.3$ (based on power to detect bias $>5\%$), the full Itô term is included).

The associated density $p(l,t)$ satisfies the Fokker-Planck equation:

$$\partial p / \partial t = \partial / \partial l [(v(t)/U(t)) l p] + (1/2) \partial^2 / \partial l^2 [\sigma^2 U(t)^2 p]$$

With a degenerating diffusion coefficient forcing collapse toward $L=0$, implemented as a Dirichlet absorbing condition at $L=0$ (with sensitivity tests using Robin boundaries, absorption rate $\kappa=0.1-1$, to model gradual collapse, comparing mass concentration via equivalence testing (TOST 90% CI within preregistered bounds or $BF_{10} < 1/3$ favoring no difference) confirming equivalence to Dirichlet under $\kappa \rightarrow \infty$ limit), with reflecting boundaries at large $|L|$ to simulate unbounded domain in finite-difference schemes. Numerical solutions (e.g., finite-difference methods, with numerical stability checked via the Courant-Friedrichs-Lewy stability condition for the chosen finite difference discretization) confirm

mass concentration for empirical validation, demonstrating convergence of the probability density toward a point mass at zero as $U(t) \rightarrow 0$.

As the biological reserve $U(t) \rightarrow 0$, two empirical signatures emerge:

1. **Drift Dominance:** The mean reversion rate $(v(t)/U(t))$ diverges as $U(t) \rightarrow 0$, forcing strong mean reversion provided that $v(t)/U(t)$ diverges as $U(t)$ approaches zero.
2. **Variance Compression:** The diffusion coefficient $\sigma U(t)$ vanishes, suppressing stochastic excursions and driving the probability distribution toward a degenerate point mass at zero as the terminal boundary is approached.

Predictions must account for sensor noise near the endpoint via Kalman filtering (with example calibration values such as process noise $Q \approx 0.01$ and measurement noise $R \approx 0.1$, subject to empirical tuning) or similar preregistered pipelines.

The system cannot know the realized T in advance. Instead, it operates on the conditional horizon: $H_t = E[T - t \mid \mathcal{F}_t]$ derived from the Lagrangian $\min J + \lambda(L(T)=0)$, the shadow price $\lambda \approx 1/H(t)$ under linear approximation, derived under the assumption of small σ where the process approximates a deterministic path, which is valid for $\sigma < 0.1$ (calibrated from HCI variance); for larger σ , we use the full Lagrangian via the Hamilton-Jacobi-Bellman equation in MDP simulations. From $E[L(T) \mid \mathcal{F}_t] = 0$, the expected required compensation scales approximately as $F_{\text{req}} \approx -L(t)/H(t)$ under a Brownian small variance assumption, with non-Markovian extensions via LSTM for $H(t)$ estimation.

Define the admissible set $A_t \subset A$ as the menu of actions available at time t that preserve the feasibility of terminal closure given the current ledger state L_t , horizon H_t , and available relief channels C . Options that render closure mathematically infeasible (e.g., if the action's projected $|F_{\text{req}}|$ exceeds a preregistered feasibility bound based on available physiological channels C) drop out of the admissible set or are removed or strongly suppressed.

Compensability $\Phi(a \mid L(t))$ quantifies the expected change in closure feasibility, formally defined as: $\Phi(a) = E[\Delta(\text{closure probability}) \mid a, L(t)]$ via preregistered Markov Decision Process (MDP) simulations. The shadow price $\lambda(t)$, formally defined as the Lagrange multiplier associated with preserving the admissibility of terminal closure, scales directly with the inverse horizon $H(t)^{-1}$. As $H(t)$ shrinks, pruning tightens, increasing the selection weight on repair and relief actions.

Empirical Horizon Scaling Predictions: Operationally, admissible menu counts should decrease $\propto H_t^{-1}$ and exhibit overdispersion rigorously tested via preregistered Negative Binomial generalized linear mixed models, with model fit compared to Poisson alternatives using AIC/BIC criteria. Neurobiologically, this manifests as a Queue System (QS) residual.

After controlling for utility, conflict, arousal, and fatigue, control hubs such as the rIFG and ACC building on established models of time-pressure threshold adjustments (Shenhav, 2013) must retain residual variance strictly explained by Φ and Horizon $\times$ Φ interaction terms. Absence of this residual signal under adequate statistical power (powered for 80% detection of residual variance $> 5\%$, $n \geq 200$) constitutes a mechanistic strike against the QS hypothesis.

In waking life, external constraints may prevent behavioral compensation (e.g., imprisonment, systemic paralysis). If LoF holds as a 100 percent guarantee, one possible biological pathway could involve internal neural dynamics capable of altering affective balance offline. One candidate biological mechanism consistent with LoF would produce a statistical relationship between waking ledger accumulation L_{wake} and subsequent REM dream valence F_{REM} : $E[\text{negative dream content} | \text{negative } L_{\text{wake}}]$ is high early in the night, decreasing overnight (Cartwright, 1998); this is a true homeostatic inversion, not merely memory consolidation.

Mechanism Framing: NREM slow-wave down-selection manages global metabolic load (via the Synaptic Homeostasis Hypothesis, SHY). Conversely, offline affective balancing operates critically during REM sleep. Healthy REM suppresses noradrenergic (norepinephrine) tone, permitting affective processing of highly valent memories without triggering the full autonomic stress response. This may contribute to affective rebalancing during sleep, phenomenologically experienced as valence-inverted dream states. This hypothesis extends emotion regulation theories (Cartwright, 1998) and aligns with state restoration frameworks (Hobson, 2009).

Empirical Falsifiers: Morning dream reports are coded for valence alongside polysomnography. The mechanism is falsified if REM valence correlation > -0.3 or TOST 90% CI within $[-0.3, 0.3]$ or $BF_{10} < 1/3$ favoring no association after controlling for recall and non-affective REM functions (e.g., standard memory consolidation), with preregistered power analysis targeting 80% power to detect correlations ≤ -0.3 (e.g., $n \approx 85$ per group assuming $\alpha = 0.05$) (intraclass correlation > 0.7 required for dream valence coding, excluding medicated cases). The prediction dictates that high negative waking accumulation, when processed by healthy, non-pathological sleep architecture, drives a homeostatic inversion, potentially manifesting as a shift toward more positively valenced or mastery-themed REM states (extending observations of decreasing negative content in Cartwright, 1998).

Biological Caveat: Pathological states such as PTSD or chronic depression involve documented REM anomalies and heightened amygdala reactivity linked to noradrenergic dysregulation (Germain, 2008). Rather than dismissing these anomalies as mere mechanistic exceptions, the framework treats them as hard boundary tests. If recurrent

nightmares prevent the necessary inversion at a population prevalence exceeding preregistered bounds of >30% imbalance in PTSD cohorts (aligned with Germain, 2008 meta-prevalence of REM anomalies ~30-50% in PTSD, measured as HCl > 2σ post-REM, controlling for amygdala hyperactivity via fMRI covariates), the mechanism is definitively falsified. While hypothesized as a modifiable vulnerability factor, bidirectional causality between PTSD and sleep disturbances is acknowledged; preregistered longitudinal designs will disentangle directions via cross-lagged models.

If $L(T)=0$ is a structural boundary condition, why does such a constraint persist biologically? Because sustained extreme affective states are metabolically costly and potentially destabilizing. Persistent high-magnitude valence (whether chronic distress or unyielding mania) corresponds to sustained prediction error and chronic autonomic activation. Maintaining such states requires continuous ATP expenditure which may contribute to increased metabolic load and accelerated depletion of biological plasticity $U(t)$.

Unbounded affective variance destabilizes the organism metabolically. Drawing on established indices of allostatic load (Juster, 2010) and models of allostatic load energetics (Bobba-Alves, 2022), high-variance trajectories systematically accelerate systemic entropy. Persistent extremes likely accelerate entropy, as per allostatic models, testable via mediation (affective variance → epigenetic age, path coefficient > 0.2, bootstrapped CI excluding 0, controlling for correlated behavioral factors like impulsivity). Correlational evidence suggests affective variance may accelerate aging (Juster, 2010); this causation is testable via preregistered MZ twin discordance in affective variance predicting epigenetic clock differences (e.g., $\Delta\text{Horvath clock} > 2$ years, $p < 0.05$, $n \geq 50$ pairs). We predict ongoing models (Bobba-Alves, 2022) will quantify this metabolic penalty, with evolution selecting indirectly for physical stability, akin to frailty models (Vaupel, 2010).

No teleology or cosmic intent is required. Systems that fail to regulate the time-integrated affective load simply exit the state space faster under metabolic constraints. Constraint binding control architectures are perfectly compatible with blind evolutionary selection pressures acting on thermodynamic stability. Limiting the time-integral of affective variance is biologically aligned with preserving complex, low-entropy organization. The neutrality boundary is not a moral reward system; it is a physical stability constraint demanded by the thermodynamics of maintaining an integrated nervous system.

Standard homeostatic models, such as Ornstein-Uhlenbeck dynamics, define affect as:

$$dF(t) = -\kappa F(t) dt + \sigma dW(t)$$

although this locally regresses to the mean, the integrated ledger $L(t) = \int_0^t F(s) ds$

accumulates variance over time, where the exact closed-form

$$\text{Var}(L(t)) = (\sigma^2 / (2\kappa^3))(2\kappa t - 3 + 4e^{-\kappa t} - e^{-2\kappa t})$$

grows asymptotically linear with slope $\sigma^2/(2\kappa)$ for large t . For HCI series, κ is calibrated from autocorrelation via AR(1) fits; for sufficiently large κ relative to the observational horizon T the variance growth slows but does not vanish, weakening the contrast to LoF's compression; test via model comparison (AIC OU - AIC LoF > 2 favoring LoF). As a result, passive random surpluses or deficits at T remain possible with non-zero probability under standard modeling.

Because LoF hypothesizes $L(T) = 0$ as a 100 percent guarantee, the process cannot remain an unconstrained diffusion. It must behave like a bridge-like constrained diffusion or controlled diffusion process ensuring strict path-wise closure. The critical empirical test emerges as the conditional horizon $H(t)$ approaches zero (e.g., in hospice contexts). A central prediction of this framework is that cross-sectional ledger variance contracts as the process approaches the terminal stopping time T under comparable reserve scaling $U(t)$, i.e.,

$$\lim_{t \rightarrow T} \text{Var}(L(t)) = 0$$

under the coupled process $L(t) = U(t)Y(t)$ (estimated via rolling variances in preregistered bins scaled to $U(t)$).

While theoretically derived from diffusion equations, this boundary condition beautifully mirrors established physiological phenomena in gerontology, specifically the "terminal drop" in autonomic variability in older adults (Kim, 2010; Lipsitz & Goldberger, 1992). Ledger variance is proxied via HCI rolling variances, with homogeneity tested via Levene's test on preregistered $U(t)$ -scaled bins to empirically confirm this contraction, explicitly controlling for confounds such as polypharmacy via propensity score matching (caliper=0.2 SD, including age, comorbidities, and ATC medication codes) or covariate adjustment for medication classes (e.g., opioids, beta-blockers). In the period preceding natural death, organisms exhibit a measurable collapse in cognitive and autonomic dynamism, observed physiologically as dramatic flattening of Heart Rate Variability (HRV), which serves as the physical substrate for affective flexibility. The organism literally loses the energetic capacity to mount extreme affective excursions. The mathematical variance compression of the ledger cleanly formalizes this terminal thermodynamic collapse.

We employ Two One-Sided Tests (TOST) to evaluate terminal neutrality.

- **Neutral (Corroborated):** The 95 percent confidence interval (95% CI, for conservatism in TOST to minimize Type I errors) of the empirical ledger estimate $\hat{L}(T)$ lies entirely within the preregistered bounds $[-K, +K]$.

- **Non-neutral (Falsified):** If the 95 percent confidence interval of the estimator $\hat{L}(T)$ lies entirely outside $[-K, +K]$, the equivalence test rejects neutrality.
- **Inconclusive:** Overlap occurs without full containment.

To ensure additional statistical rigor, TOST is supplemented with Bayes factors computed against a preregistered uninformative prior (e.g., Cauchy(0,1) for effect size). For TOST, equivalence bounds K scale with T (e.g., $K = \sigma\sqrt{T}$). To account for potential skewness in path-dependent non-Gaussian physiological sums, this is evaluated using 10,000 BCa bootstraps. For highly non-Gaussian $\hat{L}(T)$, robust Bayes with a Cauchy or t-prior (scale=0.707) is used to confirm $BF_{01} > 30$ for very strong evidence per updated conventions (Wagenmakers, 2018), preventing subjective tuning. Levene's test for variance homogeneity requires $\eta^2 > 0.06$ (small-medium effect per Cohen, 1988). A $BF_{01} > 30$ is required as very strong evidence favoring neutrality relative to the unconstrained alternative. Conversely, a $BF_{10} > 30$ in favor of terminal imbalance constitutes a definitive statistical falsifier.

LoF is refuted under any of the following hard conditions (applying Holm-Bonferroni correction across falsifiers, $\alpha=0.05$ family-wise):

- **Terminal imbalance:** repeated well-measured streams end with confidence intervals strictly outside the neutrality band (falsified if $>10\%$ of streams show CI outside $[-K, K]$, powered for 80% detection at 10% rate, e.g., $n=120$ streams, $\alpha=0.05$).
- **Variance expansion:** longitudinal data show expanding ledger variance near T (measured via Levene's test for homogeneity of variance across preregistered time bins), violating the compression signature.
- **Invariance breakdown:** if the HCI metric fails scalar or metric invariance near death, cross-group comparisons collapse and the theory loses empirical footing.

Operational Summary: The empirical ledger estimator $\hat{L}(t)$ is constructed by integrating a preregistered Hedonic Composite Index (HCI) derived from longitudinal autonomic and metabolic markers. The endpoint T is defined as a stopping time on a preregistered unity threshold $\theta(t)$, not by solving $L(T)=0$. Neutrality is tested using a preregistered $\pm K$ equivalence band with TOST and Bayes factor confirmation. Scientific integrity is preserved by freezing the endpoint definition T , the weights w_k , and the error budget K before data collection (all empirical studies naturally require IRB approval, with strict ethical safeguards for vulnerable populations, such as informed consent proxies in hospice contexts). Under these constraints, the Law of Fairness stands as a vulnerable, testable scientific hypothesis, not a metaphysical safeguard. The 100 percent guarantee earns its scientific validity precisely because it is experimentally breakable. The theoretical scaffolding is locked; the next step is falsification.

References

- Bäckman, L., Lindenberger, U., Li, S. C., & Nyberg, L.** (2010). Linking cognitive aging to alterations in dopamine neurotransmitter functioning: recent data and future avenues. *Neuroscience & Biobehavioral Reviews*, 34(4), 670-677.
- Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D.** (2001). Bad is stronger than good. *Review of General Psychology*, 5(4), 323-370.
- Bobba-Alves, N., Juster, R. P., & Picard, M.** (2022). The energetic cost of allostasis and allostatic load. *Psychoneuroendocrinology*, 146, 105951.
- Cartwright, R., Young, M. A., Mercer, P., & Bears, M.** (1998). Role of REM sleep and dream affect in overnight mood regulation: a study of normal volunteers. *Psychiatry Research*, 81(1), 1-8.
- Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., ... & Massimini, M.** (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198), 198ra105.
- Chida, Y., & Steptoe, A.** (2009). Cortisol awakening response and psychosocial factors: a systematic review and meta-analysis. *Psychological Medicine*, 39(3), 399-410.
- Cohen, J.** (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Comolatti, R., Casali, A. G., Casarotto, S., Halder, S., Laureys, S., ... & Massimini, M.** (2019). A fast and general method to empirically estimate the complexity of brain responses to transcranial and intracranial stimulations. *Brain Stimulation*, 12(5), 1180-1189.
- Garcia, J., & Fernández, F.** (2015). A comprehensive survey on safe reinforcement learning. *Journal of Machine Learning Research*, 16(1), 1437-1480.
- Germain, A., Buysse, D. J., & Nofzinger, E.** (2008). Sleep-specific mechanisms underlying posttraumatic stress disorder: integrative review and neurobiological hypotheses. *Sleep Medicine Reviews*, 12(2), 185-195.
- Gottman, J. M.** (1994). *What Predicts Divorce? The Relationship Between Marital Processes and Marital Outcomes*. Lawrence Erlbaum Associates.
- Gottman, J. M.** (1999). *The Marriage Clinic: A Scientifically Based Marital Therapy*. W.W. Norton & Company.
- Hesp, C., Smith, R., Parr, T., Allen, M., Friston, K. J., & Ramstead, M. J.** (2021). Deeply felt affect: The emergence of valence in deep active inference. *Neural Computation*, 33(3), 598-646.
- Hobson, J. A.** (2009). REM sleep and dreaming: towards a theory of protoconsciousness. *Nature Reviews Neuroscience*, 10(11), 803-814.

Hofstede, G., Hofstede, G. J., & Minkov, M. (2010). *Cultures and Organizations: Software of the Mind* (3rd ed.). McGraw-Hill.

Horvath, S. (2013). DNA methylation age of human tissues and cell types. *Genome Biology*, 14(10), R115.

Juster, R. P., McEwen, B. S., & Lupien, S. J. (2010). Allostatic load biomarkers of chronic stress and impact on health and cognition. *Neuroscience & Biobehavioral Reviews*, 35(1), 2-16.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263-291.

Kim, D. H., Lipsitz, L. A., Ferrucci, L., Varadhan, R., Guralnik, J. M., ... & Lipsitz, L. A. (2010). Association between reduced heart rate variability and frailty. *Journal of the American Geriatrics Society*, 58(8), 1698-1703.

Kraskov, A., Stögbauer, H., & Grassberger, P. (2004). Estimating mutual information. *Physical Review E*, 69(6), 066138.

Li, S. C., & Sikström, S. (2002). Integrative neurocomputational perspectives on cognitive aging, neuromodulation, and representation. *Neuroscience & Biobehavioral Reviews*, 26(7), 795-808.

Lipsitz, L. A., & Goldberger, A. L. (1992). Loss of 'complexity' and aging. Potential applications of fractals and chaos theory to senescence. *JAMA*, 267(13), 1806-1809.

McEwen, B. S., & Wingfield, J. C. (2010). What is in a name? Integrating homeostasis, allostasis and stress. *Hormones and Behavior*, 57(2), 105-111.

Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1-2), 125-144.

Shenhav, A., Botvinick, M. M., & Cohen, J. D. (2013). The expected value of control: an integrative theory of anterior cingulate cortex function. *Neuron*, 79(2), 217-240.

Smith, R., Badcock, P., & Friston, K. J. (2019). Recent advances in the application of predictive coding and active inference models within clinical neuroscience. *Psychiatry and Clinical Neurosciences*, 75(1), 3-13.

Vaupel, J. W. (2010). Biodemography of human ageing. *Nature*, 464(7288), 536-542.

Wagenmakers, E. J., Marsman, M., Jamil, T., Ly, A., Verhagen, J., Love, J., ... & Morey, R. D. (2018). Bayesian inference for psychology. Part I: Theoretical advantages and practical ramifications. *Psychonomic Bulletin & Review*, 25(1), 35-57.